

Table S1: The 35 implemented tasks: domain, inspired paradigm, principal outputs, difficulty mechanism, software verification, and psychometric status. Scoring fixture (F) = hand-derived numerical fixture for the task’s scoring function in the frontend Vitest suite; Scoring fixture (B1–B4) = hand-derived numerical fixture in backend pytest batch 1–4; “+ behavioural tests” = additional tests of trial generation, response classification, and difficulty adjustment. A fixture checks a specific calculation, not the validity of the task. NE = not established (no reliability, validity, or normative data). Levels are developer-specified; no patients with multiple sclerosis were evaluated.

| Task                        | Domain           | Inspired paradigm                                      | Principal outputs                                     | Difficulty / adaptation                                                                                       | Software verification                    | Psychometric status |
|-----------------------------|------------------|--------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------|
| N-Back                      | Working memory   | Visual $n$ -back updating [1]                          | accuracy, mean RT, misses, false alarms               | Baseline: 20 trials at 1-back; training mode uses 1–3-back by level. Dual N-Back is a separate training task  | Scoring fixture (F)                      | NE                  |
| Digit Span                  | Working memory   | Digit recall (forward, backward, mixed) [2]            | trial accuracy, span, score                           | Levels 1–10: length 3–9 and forward/backward/mixed recall; trials 4 and 7 use one level higher                | Scoring fixture (B1)                     | NE                  |
| Spatial Span                | Working memory   | Spatial-sequence recall (Corsi-type) [3]               | set and order accuracy, maximum span, score           | Levels 1–10: sequence length, display and interval timing                                                     | Scoring fixture (B2)                     | NE                  |
| Letter–Number Sequencing    | Working memory   | Ordered manipulation of mixed letters and digits [2]   | set accuracy, order accuracy, speed, score            | Levels 1–10: sequence composition and length, timing                                                          | Scoring fixture (B2)                     | NE                  |
| Operation Span              | Working memory   | Complex span (arithmetic check plus letter recall) [4] | arithmetic accuracy, letter accuracy, dual-task score | Levels 1–10: set size, operation complexity, timing                                                           | Scoring fixture (B2)                     | NE                  |
| Dual N-Back                 | Working memory   | Dual-modality (position and letter) $n$ -back [5]      | visual, auditory and dual accuracy, score             | Levels 1–10: $n = 1–3$ , stream length, timing, target rate                                                   | Scoring fixture (B4) + behavioural tests | NE                  |
| Simple Reaction Time        | Processing speed | Simple reaction time [6]                               | mean RT, RT SD, choice accuracy, combined score       | Shared processing-speed workflow; levels 1–10 change delay, alternatives, timing                              | Scoring fixture (F)                      | NE                  |
| Symbol–Digit Matching       | Processing speed | Symbol–digit substitution [7]                          | correct, incorrect, completion time, score            | Levels 1–10: symbol count, item count, exposure, target                                                       | Scoring fixture (B1)                     | NE                  |
| Numeric Trail Task          | Processing speed | Sequential visual trail [8]                            | completion time, errors, score                        | Levels 1–10: node count, layout, expected time                                                                | Scoring fixture (B2)                     | NE                  |
| Pattern Comparison          | Processing speed | Rapid same/different comparison [9]                    | accuracy, RT, timeouts, score                         | Levels 1–10: grid size, similarity, changed cells, duration, trials                                           | Scoring fixture (B1)                     | NE                  |
| Inspection Time             | Processing speed | Brief line discrimination with mask [10]               | accuracy, threshold duration, score                   | Levels 1–10: presentation time, line difference, trials                                                       | Scoring fixture (B1)                     | NE                  |
| Choice Reaction Time        | Processing speed | Stimulus–response choice RT [11]                       | accuracy, mean RT, consistency, score                 | Levels 1–10: 2–4 alternatives, trials, windows; up if score $\geq 83$ and accuracy $\geq 82$ , down if $< 63$ | Scoring fixture (B4) + behavioural tests | NE                  |
| Continuous Performance Task | Attention        | Sustained target detection (AX-type baseline) [12]     | target hit rate, false clicks, mean RT, score         | Baseline: 60 trials; training mode adds 10 trials per level above 3. Go/No-Go is a separate training task     | Scoring fixture (F)                      | NE                  |

| Task                         | Domain          | Inspired paradigm                            | Principal outputs                                            | Difficulty / adaptation                                                                                                                    | Software verification                    | Psychometric status |
|------------------------------|-----------------|----------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------|
| Serial Addition              | Attention       | Serial addition (PASAT-type) [13]            | correct, omissions, RT, accuracy-based score                 | Levels 1–10: shorter inter-stimulus interval, trials                                                                                       | Scoring fixture (B1)                     | NE                  |
| Color–Word Interference      | Attention       | Color–word interference [14]                 | accuracy, interference score, RT, score                      | Levels 1–10: color set, congruency ratio, timing, trials                                                                                   | Scoring fixture (B3)                     | NE                  |
| Go/No-Go                     | Attention       | Response inhibition [30, 31]                 | go and no-go accuracy, RT, consistency, score                | Levels 1–10: stimulus similarity, no-go frequency, timing, trials                                                                          | Scoring fixture (B3)                     | NE                  |
| Flanker                      | Attention       | Flanker interference [15]                    | accuracy, congruent/incongruent RT, conflict control, score  | Levels 1–10: congruency ratio, spacing, timing, trials                                                                                     | Scoring fixture (B3)                     | NE                  |
| Sustained-Attention Response | Attention       | Frequent-go, rare-no-go vigilance [16]       | overall and no-go accuracy, RT, vigilance penalty, score     | Levels 1–10: stream length, target rate, timing; up if score $\geq 84$ and $\leq 4$ commissions, down if $< 64$                            | Scoring fixture (B4) + behavioural tests | NE                  |
| Task Switching               | Flexibility     | Cued task switching [17]                     | accuracy, switch cost, perseveration errors, score           | Baseline: 40 trials; training mode adds 10 trials per level above 4. Rule Shift is a separate training task                                | Scoring fixture (F)                      | NE                  |
| Alternating Trail Task       | Flexibility     | Alternating number–letter trail [8]          | completion time, errors, B-minus-A                           | Levels 1–10: node count, layout, expected time                                                                                             | Scoring fixture (B3)                     | NE                  |
| Card-Sorting Rule Shift      | Flexibility     | Feedback-guided card sorting [18]            | categories, perseverative errors, accuracy, score            | Levels 1–10: trials, rule-change frequency, feedback, category complexity                                                                  | Scoring fixture (B3)                     | NE                  |
| Dimensional Card Sort        | Flexibility     | Dimensional rule switching [19]              | accuracy, mean RT, switch cost, score                        | Levels 1–10: dimensions, trials, switch frequency, timing                                                                                  | Scoring fixture (B3)                     | NE                  |
| Rule Shift                   | Flexibility     | Cued set shifting (color, shape, count) [17] | overall and switch accuracy, RT, perseverative errors, score | Levels 1–10: block sequence, cue time; up if score $\geq 83$ , switch accuracy $\geq 72$ and $\leq 5$ perseverative errors, down if $< 63$ | Scoring fixture (B4) + behavioural tests | NE                  |
| Plus–Minus Switching         | Flexibility     | Alternating arithmetic switching [20]        | accuracy, mean RT, switch cost, score                        | Levels 1–10: operand range, mixed-block frequency, items, timing                                                                           | Scoring fixture (B3)                     | NE                  |
| Tower Planning               | Planning        | Tower-type planning [21]                     | problems solved, moves, planning efficiency, time, score     | Levels 1–10: problem count, disks, minimum moves, time limits                                                                              | Scoring fixture (B3)                     | NE                  |
| Stockings-Style Planning     | Planning        | Tower-style planning with balls [22]         | problems solved, planning efficiency, mean time, score       | Levels 1–10: problems, balls, minimum moves, time limits                                                                                   | Scoring fixture (B3)                     | NE                  |
| Letter Fluency               | Planning        | Phonemic fluency [23]                        | valid unique words, errors, mean words, score                | Levels 1–10: letter sets, prompt count, duration                                                                                           | Scoring fixture (B3)                     | NE                  |
| Category Fluency             | Planning        | Semantic fluency [23]                        | unique valid words, duplicates, score                        | Levels 1–10: category pools; expected count rises with level                                                                               | Scoring fixture (B4) + behavioural tests | NE                  |
| Twenty Questions             | Planning        | Constraint-seeking problem solving [24]      | identification, questions used, strategy score, score        | Levels 1–10: five object pools in rotation; score multiplier $1 + 0.05(\text{level} - 1)$                                                  | Scoring fixture (B4) + behavioural tests | NE                  |
| Visual Search                | Visual scanning | Target search among distractors [25]         | targets found, accuracy, time per target, score              | Levels 1–10: item count, distractor similarity, targets, grid                                                                              | Scoring fixture (B4) + behavioural tests | NE                  |

| Task                           | Domain          | Inspired paradigm                         | Principal outputs                                    | Difficulty / adaptation                                                                                                                 | Software verification                    | Psychometric status |
|--------------------------------|-----------------|-------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------|
| Cancellation                   | Visual scanning | Letter or symbol cancellation [26]        | hits, omissions, commissions, time, score            | Levels 1–10: 150–540 cells, target count and types                                                                                      | Scoring fixture (B4) + behavioural tests | NE                  |
| Feature and Conjunction Search | Visual scanning | Feature-integration search [25]           | accuracy, search time, set-size slope                | Levels 1–10: set size, similarity, conjunction load                                                                                     | Scoring fixture (B3)                     | NE                  |
| Landmark Line Judgment         | Visual scanning | Line-bisection judgment [27]              | accuracy, omissions, spatial bias, RT, score         | Levels 1–10: offset distribution, exposure; up if score $\geq 84$ , $ \text{bias}  \leq 8$ and $\leq 1$ omission, down if $< 64$        | Scoring fixture (B4) + behavioural tests | NE                  |
| Multiple-Object Tracking       | Visual scanning | Dynamic object tracking [28]              | accuracy, precision, recall, F1, score               | Levels 1–10: 6–16 objects, 1–3 targets, speed, duration                                                                                 | Scoring fixture (B4) + behavioural tests | NE                  |
| Central–Peripheral Attention   | Visual scanning | Divided central/peripheral attention [29] | central accuracy, peripheral localization, RT, score | Levels 1–10 shorten presentation time (800 to 150 ms): levels 1–3 central only, 4–6 add a peripheral target, 7–10 add 24–47 distractors | Scoring fixture (B2)                     | NE                  |

**Note.** N-Back, Continuous Performance Task, Task Switching, and Simple Reaction Time are scored client-side in the baseline workflow. Their scoring functions are extracted to a frontend module and each is covered by one Vitest fixture (F); the other 31 tasks are scored by backend services and covered by pytest fixtures (B1–B4).

## References

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